

Final Project:

Documentation

Final Project: Instrument/Sensor Design

Temperature Data Acquisition System

Shadrack Agyei Nti- 49792026

Sean Nana Boateng Yeboah - 10952026

ENGR212: Instrumentation for Engineering

Kofi Adu Larbi

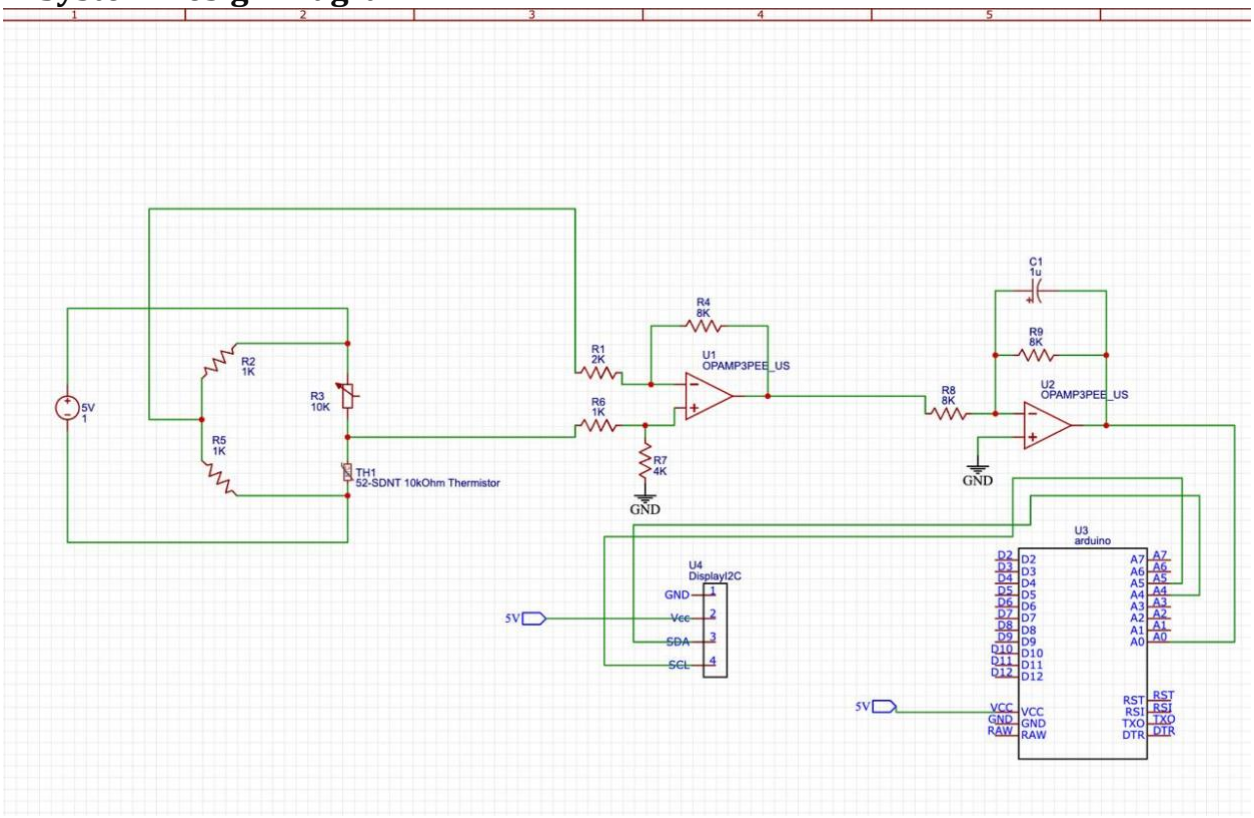
December 13th, 2024

Technical Documentation: Temperature Measurement System Design

Project Overview:

This system is designed to measure temperature within the range of **25°C to 35°C**, with a **resolution of 0.01°C**. The system utilizes an **NTC thermistor** for temperature sensing, an **ADC (Analog to Digital Converter)** for digitizing the signal, and an **LCD** for displaying the temperature readings. The system is calibrated using a **lookup table** for accurate temperature-resistance mapping, ensuring precision in measurements. The system is suitable for use in Fab Lab projects requiring precise thermal monitoring within a low-frequency range of **0–20 Hz**.

1. System Design Diagram:



2. Subsystem Design Decisions:

Sensing Subsystem:

- **Component Used:** NTC thermistor (Negative Temperature Coefficient).
- **Reasoning:** NTC thermistors provide a nonlinear resistance-temperature relationship, making them ideal for precise temperature measurements within the target range.

Signal Conditioning Subsystem:

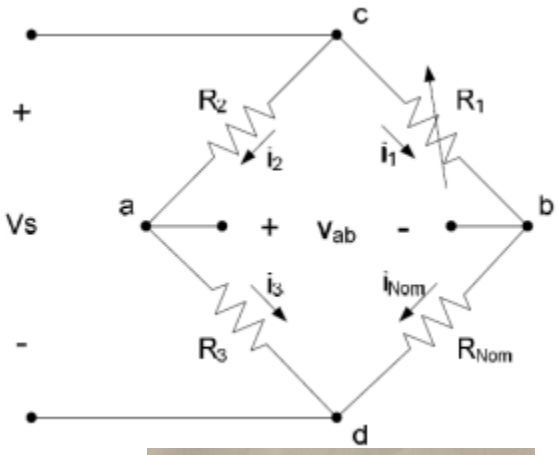
Wheatstone Bridge**:** Converts the resistance change of the NTC thermistor into a voltage change.

Difference Amplifier**:** Amplifies the small voltage difference from the Wheatstone bridge for better ADC resolution.

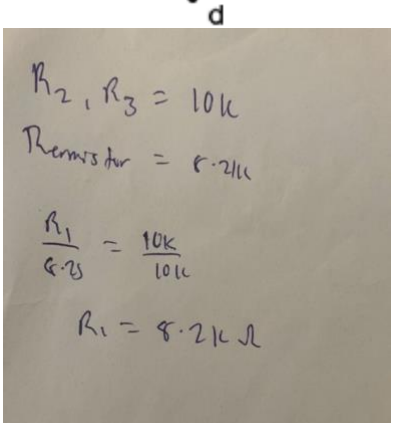
Active Low-Pass Filter**:** Filters out high-frequency noise above 20 Hz, ensuring stable signal conditioning.

Design Steps:

1. **Wheatstone Bridge:**



- The Wheatstone bridge circuit was designed as shown in the figure.
- The bridge was roughly balanced by adjusting the potentiometer to achieve zero voltage output at room temperature (25°C).
- The Wheatstone bridge was designed to convert the resistance variation of the thermistor into voltage variation. At room temperature (25°C), the output voltage should be approximately 0V. The balancing of the bridge was done using a potentiometer, and we chose a 10K potentiometer to get better accuracy.
- For a balanced bridge, the voltage difference v_{ab} is zero.
- We used 10k resistors to limit the current through the circuit



2. **Difference Amplifier:**

A difference amplifier is used to amplify the small differential output voltage of the Wheatstone bridge.

The amplifier gain is set to ensure that the output voltage matches the input range of the ADC.

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

$$v_{out} = \frac{R_2}{R_1}(v_b - v_a)$$

R1,R3 = 1K

R2,R4 = 10K

The output voltage of the difference amplifier at room temperature: 0.4V

The output voltage of the difference amplifier at body temperature: 3.9V

Gain = Vout/Vin

= 3.9/0.4

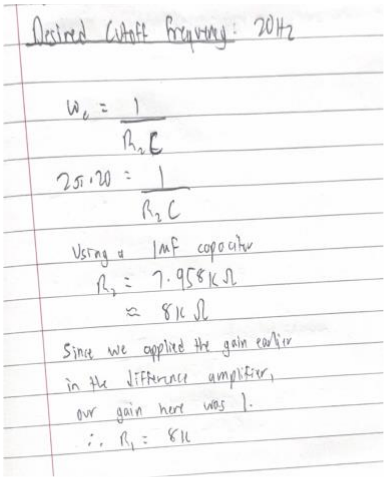
= 10.

3. **Active Low-Pass Filter:**

An operational amplifier-based RC circuit is used as the active low-pass filter.

Cut-off frequency (f_c) is designed at 20 Hz to remove noise while preserving the signal of interest:

The filter ensures that only the temperature signal (0–20 Hz) is passed to the ADC.



Signal Conversion and Processing Subsystem:

The ADC of our Arduino Uno converts the analog voltage from the thermistor into a digital number. The microcontroller uses the digital value as an index to the lookup table. It retrieves the corresponding temperature for that ADC value, which is then used to display the temperature.

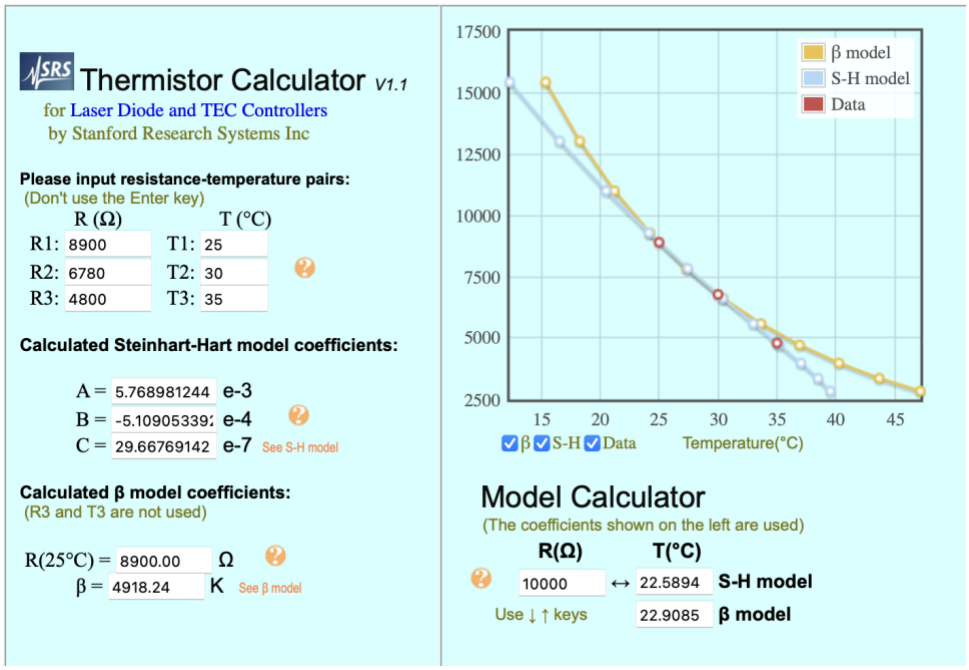
- **Lookup Table Generation:**
  - Generated using the Steinhart-Hart equation.
  - Covers the temperature range (25°C to 35°C) with intervals corresponding to 0.01°C resolution.

$$\frac{1}{T} = A + B\ln(R) + C(\ln(R))^3$$

Where:

- T is the temperature in kelvins (K),
- R is the thermistor's resistance in ohms (Ω),
- A , B , and C are the Steinhart–Hart coefficients specific to the thermistor.

We measured and recorded the thermistors resistance at temperatures (25,30 and 35) and using the thermistor calculator we got our coefficients.



Display Subsystem:

- **Component Used:** 16x2 LCD module.
- **Interfacing:** Connected to the microcontroller via I2C

4. Results and Conclusion

The system accurately measures temperature within the range of **25°C to 35°C**.

Achieved a resolution of **0.01°C**, meeting the project requirements.

The use of the pre-calibrated lookup table ensured precise conversion of ADC values to temperature readings.

The LCD successfully displayed real-time temperature values, with clear readability and responsiveness.

Link to code:

https://github.com/ShadrackAgyei/Temperature-Measurement-System/blob/main/instrumentation_final